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Sixth Semester B.E. Degree Examination, June/July 2019
Compiler Design

Time: 3 hrs. Max. Marks:100

Note: ***Answer any FIVE full questions, selecting atleast TWO questions from each part.***

PART - A

1 a. Write a brief note on Language Processing System. (04 Marks)

b. Explain with a neat diagram, the phases of compiler. (10 Marks)

c. Construct the transition diagram to recognize the tokens given below :

i) Relational operation ii) Unsigned number. (06 Marks)

2 a. Define Left recursion and Left factoring and apply the same for the Grammar

$E \rightarrow E * T / T$; $T \rightarrow id + T / id$ (06 Marks)

b. Given the Grammar

$S \rightarrow XS \mid dS \mid \in$

$X \rightarrow Y \mid Zb \mid aY$

$Y \rightarrow cZ$

$Z \rightarrow e$

i) Construct FIRST and FOLLOW sets.

ii) Construct the Predictive parsing table.

iii) Show the moves made by the predictive parser on the input "dace". (10 Marks)

c. How to verify whether grammar is LL(1) or not, show that

$S \rightarrow |AB| \in$

$A \rightarrow |AC|OC$

$B \rightarrow OS$

$C \rightarrow 1$ is LL(1) without constructing any table. (04 Marks)

3 a. What is handle pruning? Explain with the help of the grammar

$S \rightarrow (L) \mid a$

$L \rightarrow L, S \mid S$ and input string (a , (a, a)). (04 Marks)

b. Explain the conflicts that may occur during shift reduce parsing . consider dangling – else grammar. (04 Marks)

c. Given the grammar

$S \rightarrow (S) S \mid \in$ or can be written as

$S \rightarrow (S) S$

$S \rightarrow \in$

i) Find LR(0) items

ii) Construct SLR(1) parsing table and show the parsing steps for the input ( ) ( ) $. (12 Marks)

4 a. Given the grammar $S \rightarrow CC$; $C \rightarrow cC|d$

i) Construct sets of LR(1) items.

ii) Construct Canonical LR(1) Parsing table. (14 Marks)

b. Write the face specification of a simple desk calculator with the following grammar for arithmetic expressions

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid digit.$ (06 Marks)

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PART - B

5 a. Write annotated parse tree for 3 * 5 + 4n using top down approach. Write semantic rules for each. **(08 Marks)**

b. Write a brief note on dependency graph. **(04 Marks)**

c. Construct a dependency graph for the declaration float id_1 , id_2 , id_3 , **(08 Marks)**

6 a. What is DAG? Construct a DAG for the following expression

a + a * (b – c) + (b – c) * d **(05 Marks)**

b. Write annotated parse tree for C + a [i] [j] and derive 3 – address code for the same expression. **(08 Marks)**

c. Write S.D translation for Switch statement. **(07 Marks)**

7 a. Explain run – time storage scheme for C++ language. Give the structure of activation record and explain the purpose of each item. **(10 Marks)**

b. What are access links? Explain how are access links determined for finding non local data, what is its drawback. **(06 Marks)**

c. Discuss the performance metrics to be considered while designing a garbage collector. **(04 Marks)**

8 a. Discuss the issues in the design of a code - generator. **(10 Marks)**

b. Apply the Code – Generation algorithm to translate the basic block shown below

t = a – b
u = a – c
v = t + u
a = d
d = v + u

Assume t , u , v are temporaries , local to the block while a , b , c , d are variables that are line on exit from the block. **(10 Marks)**

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